

# Sami Koneru-Ansari

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## EDUCATION

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### University of California, Berkeley

Bachelor of Science, Electrical Engineering and Computer Science

Berkeley, CA

- **GPA: 4.0**
- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Machine Structures, Signal Processing, Computer Networks, Discrete Mathematics, Probability Theory, Linear Algebra

## WORK EXPERIENCE

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### DreamIt

Feb. 2026 – Present

Software Engineer

Berkeley, CA

- Directing a **3-person engineering team** to architect and ship end-to-end Rust features for an early-stage no-code gaming app, scaling the platform to support **1,400+ users** and public game deployments.
- Architected an async Claude API pipeline for end-to-end game generation from prompts in **under 2 minutes**.

### TokenWorks

Feb. 2026 – Present

Machine Learning Research Intern

Remote

- Fine-tuning open-source models such as the ByT5 transformer to detect and translate low-resource languages in **over 200,000 OCR-processed documents** using aligned corpora from Wikidata and previous OCR runs.
- Building a containerized Wikibase with Docker to serve a **1,400-item** knowledge graph of the Sumerian lexicon.

### GetGreen

Sept. 2025 – Dec. 2025

Machine Learning Engineer (Contract)

Remote

- Engineered a **Retrieval-Augmented Generation** chatbot using LangChain for LLM orchestration and ChromaDB for vector storage and semantic search to deliver environmental answers in **under 3 seconds**.
- Scraped and processed **1,000+ articles** with BeautifulSoup, Pandas, and SQL for a model retrieval corpus.

### California Department of Technology

May 2024 – July 2024

Software Development Intern

Rancho Cordova, CA

- Developed full-stack web apps with HTML/CSS/JS for frontend, Node.js for backend, PostgreSQL for relational data, and JWT authentication to manage scheduling and file sharing for **100+ employees**.
- Managed datasets of **500+ employees** and **70+ clients** using SQL and Excel to drive operational reporting.

## PROJECTS

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### Custom Digit Classifier

C++, CUDA

- Engineered and trained a **Convolutional Neural Network from scratch** in C++/CUDA, hand-writing a training and backpropagation pipeline for convolution, pooling, and dense layers with Adam optimization.
- Wrote GPU-optimized CUDA kernels implementing im2col and GEMM, reaching **over 99% accuracy** on the MNIST dataset and running at **~10% the throughput** of an equivalent PyTorch (cuDNN) model on T4 GPUs.

### GPT-2

Python, PyTorch, CUDA

- Implemented an efficient **124M parameter GPT-2** transformer model based on the decoder architecture from the **“Attention Is All You Need”** research paper with causal self-attention and positional/token encodings.
- Optimized training for **CUDA-enabled GPUs**, including multi-GPU execution, efficient tensor operations, and minimized CPU-GPU synchronization overhead, resulting in **an 11x speedup** from the pre-optimized model.

## TECHNICAL SKILLS

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- **Languages:** Python, C++, Java, C#, SQL, Rust, JavaScript, TypeScript, HTML/CSS, Scheme, Bash
- **Libraries:** NumPy, Pandas, PyTorch, Matplotlib, React, Node.js, BeautifulSoup, Scikit-learn, Flask, LangChain
- **Tools:** Git, Docker, Kubernetes, AWS, Ollama, Linux, CUDA, PostgreSQL, ChromaDB, Supabase